

Yuri Bochkarev

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Summary

Linux backend developer with 15+ years of experience in building, testing, and maintaining complex software systems. Over 6 years of experience with Python, 5+ years with Go, and 6+ years with C++, along with a low-level programming background. Interested in the design and architecture of cloud services, distributed systems, algorithms, and data structures.

Experience

reMarkable – paper tablets

Oslo

Senior Software Developer (Cloud Service)

December 2024 – present

- Developed a prototype / PoC of a next-generation synchronization protocol and service (Golang, HTTP, WebSockets, PostgreSQL, Spanner)
- Keywords: GCP/Google Cloud, Grafana

Pexip – video conferencing solution

Oslo

Senior Software Developer (Cloud Service)

June 2019 – December 2024

- Developed a video conferencing analytics platform that includes the following services:
- A service to track instantaneous information about calls, conferences, devices, users, etc. (Golang, HTTP, WebSockets, Google PubSub)
- A service to track historical information about the number of calls, conferences, and participants (Golang, SQLite, LMDB (embedded key-value store), ClickHouse, BigQuery, HTTP)
- A service to track and link detailed information about calls and events that constitute such calls (Golang, MySQL, Google PubSub)
- Each service has a lightweight frontend built with vanilla JavaScript
- A notification service for desktop and mobile (Golang, FCM, APN)
- Keywords: Event Sourcing, Docker, GCP/Google Cloud, Kubernetes, GitLab, Grafana, Kibana, Loki, Ubuntu

IPONWEB – media trading platform (RTB, Ad exchange, DSP, SSP)

Moscow

Senior Software Developer (Internal Tools, Bidswitch)

June 2017 – September 2019

- Transitioned the user management and access control subsystem across departments, allowing for faster and simpler workflow and operations (Python 3.6, Django 1.11, Django Rest Framework 3.6, Gabbi HTTP testing suite)
- Integrated third-party APIs with the internal discrepancy monitoring service (Python 3.6, PostgreSQL 9.6)
- Added support for bulk operations to the internal creative blocking service, saving about 1 hour per day for the client support team (Python 3.6, PostgreSQL 9.6)
- Integrated third-party API with the internal creative approval service (Python 3.5, asyncio, Cassandra 2.6)
- Optimized the user group update microservice, reducing processing time by 3x (Python 3.4, pytest, pymongo, TokuMX)
- Developed a facade API service for the internal database as part of a client integration (Golang 1.8, Cassandra 2.6)
- Added metrics to the "logs to DB" loading tool, providing better understanding and visibility of the tool's internal state (C++ 11, Cassandra 2.6)
- Software development process: Scrum, Kanban
- Keywords: Docker, Kubernetes, Sentry, Jenkins, Artifactory, OBS, Graphite, Grafana, Kibana, Ubuntu, Alpine

Senior Software Developer (Technology)

May 2016 – June 2017

- Extended the functionality of the campaign visualization and forecasting service (Python, Django, Numpy, Scipy, Pandas)
- Integrated forecasting service functionality with custom testing stands
- Software development process: Scrum + Kanban

Software Developer (Technology)

July 2014 – May 2016

- Optimized the network communication subsystem of the budget control server for advertising campaigns, reducing total iteration time by 30% (Python, Twisted, Numpy, Pandas)
- Rolled out Continuous Integration infrastructure (Jenkins, Trial, pytest, custom testing stands)

Software Developer (R&D)

January 2013 – July 2014

- Refactored and developed the data delivery service, making it more modular, testable, and robust (Python, BASH, Django)
- Added performance/time metrics to comply with the established SLA
- Standardized data transfer configuration DSL used across the company services
- Optimized file transfer time by integrating UDP-based file transfer solutions (UDT)

SmartLabs – interactive digital television (IPTV, DVB, OTT TV)

Moscow

System Department Developer

September 2009 – January 2013

- Developed an RTSP request redirector (C++, Qt)
- Developed a video content distribution and publishing automation tool (C++, Qt)

- Integrated a third-party encryption service with the video content distribution automation tool (VeriMatrix)
- Improved SmartMedia platform video server seek time by implementing a faster file segment lookup algorithm (C++, Qt, STL, live555)
- Implemented an adaptive streaming client for use in Set-Top Boxes (C++, Qt, Apple HTTP Streaming)
- Developed an interactive build automation tool that helped developers save time and make fewer mistakes when building platform components (Python)
- Contributed to the implementation of the adaptive streaming server (Python, Twisted)
- Keywords: HTTP, XML RPC, SOAP, RTSP, HLS

Autoscan

Ulyanovsk

Software Developer

September 2007 – May 2009

- Prototyped and developed network applications for real-time video transmission (C++, Qt, OPAL, libjingle)
- Automated application testing (AutoIt, Erlang)
- Keywords: XMPP, SIP, RTP, STUN

Ulyanovsk State Technical University

Ulyanovsk

Technician

September 2006 – September 2007

- computer class admin (11 computers, ASP Linux 9.0, Microsoft Windows 2003)
- set up video surveillance in the class room, technical support

Education

Ulyanovsk State Technical University

Ulyanovsk

Master's degree, Faculty of Information Systems & Technologies, diploma VMA 0105774

2007 – 2009

Bachelor's degree, Faculty of Information Systems & Technologies, diploma AVB 0512119

2003 – 2007

Additional Education

Bouvet

[coursera.org](https://www.coursera.org)

Implementing Modern Architecture June 2023

Machine Learning: Classification

April 2016

Applying DDD and EventStorming in Modern Architecture

Game Theory (HSE)

March 2016

September 2023

Machine Learning: Regression

February 2016

Information Technology Alliance

ML Foundations: A Case Study Approach

February 2016

Clojure Course

April 2013

Cloud Computing Applications

October 2015

udacity.com

Image and Video Processing

March 2015

Programming a Robotic Car

April 2012

Algorithmic Thinking

November 2014

Artificial Intelligence

December 2011

Cryptography, Part I

June 2014

Write Like Mozart

March 2014

coursera.org

Discrete Optimization

August 2013

Computing for Data Analysis

October 2012

Functional Programming Principles in Scala

December 2012

Algorithms: Design and Analysis, Part I

April 2012

Machine Learning

December 2011

Skills

Technical OS: GNU/Linux (since 2009), Windows (2001 – 2009). Programming Languages: Python, C++, Go. Moderately experienced as well in: Haskell, Clojure, R, Scala, Erlang, Lua, AutoIt, Java, Octave, x86 assembler. VCS: hg, git, svn. Technologies: Unit Testing, Design Patterns, UML. Frameworks: Twisted, Django, Pandas, Scipy.

Language Norwegian (*upper intermediate*), English (*advanced*), Russian (*native*).